The food model: specification (version 3, M98)

Version 3 restates version 2 in standard mathematical notation. No variable, parameter or equation changed (change log, section 8). The code (`plan/tools/food_model.py`) implements equations (1)-(30) and Algorithm 1. A full worked example in the same notation is in `food_model_worked_example.md`.

0. Conventions

Convention	Rule here	Basis
Order of declaration	sets and indices, then parameters (given data), then variables, then equations, then the procedure. Nothing is used before it is declared	the algebraic modelling layout of the GAMS tutorial (indices, given data, decision variables, constraints)
Symbols	one Latin or Greek letter per quantity, with sub- and superscripts; italic in the formulas	ISO/IEC 80000: quantity symbols are single italic letters
Legend	every symbol has a name, a domain (the set it is indexed over), a unit and a source	GAMS: each entity has a domain and a description; units stated with every quantity
Procedure	an algorithm with numbered lines; ← is assignment	the usual pseudocode form
Units	in square brackets; `d` = day, `t` = tonne, `p` = person	

1. The model as a map

For one city and one year:

$$\Phi:\ (N,\ \mathbf{S},\ \mathbf{w}) \longmapsto \big(\mathbf{q},\ \mathit{fed},\ x,\ \mathbf{c},\ W^{\text{plain}},\ \mathbf{L},\ \mathbf{D}\big)$$

Inputs:

- the people N;
- the stocks $\mathbf{S} = (S_g)$;
- the shares of people in each kind of work $\mathbf{w} = (w_k)$.

Outputs:

- grams eaten per person a day $\mathbf{q}$;
- how fed the people are, fed;
- energy sufficiency x;
- nutrient coverage $\mathbf{c}$;
- plain water still needed W^{plain};
- stock left $\mathbf{L}$ and spoiled $\mathbf{D}$.

The model has five modules: availability (M1), need and bounds (M2), indicators per good (M3), allocation (M4), outcomes (M5).

2. Sets and indices

Symbol	Meaning	Members
$\mathcal{G}$, index g	food and drink goods	the branch's food goods (nutrition.json)
$\mathcal{G}^{\mathrm{A}} \subset \mathcal{G}$	animal goods	Cow, Pig, Sheep, Deer, Horse, Fish, Clam, Crab, Shrimp, Whale, Dairy
$\mathcal{G}^{\mathrm{D}} \subset \mathcal{G}$	drinks	Dairy, Wine, Alcohol, Tea, Coffee
$\mathcal{G}^{\mathrm{F}} \subset \mathcal{G}$	fruit	Apple, Banana, Lemon
$\mathcal{G}^{\mathrm{T}} \subset \mathcal{G}$	brewed drinks (no energy)	Tea, Coffee
$\mathcal{G}^{\mathrm{E}} = \mathcal{G} \setminus (\mathcal{G}^{\mathrm{T}} \cup \{\mathrm{Salt}\})$	goods eaten for energy	
$\mathcal{I}$, index i	nutrients, $\lvert\mathcal{I}\rvert = 13$	protein, vitamin C, vitamin A, thiamin, niacin, folate, B12, iron, zinc, calcium, iodine, potassium, sodium
$\mathcal{K}$, index k	kinds of work	vigorous, active, sedentary
$\mathcal{P}$, index ρ	passes of the allocation	1 (nutrition), 2 (well-being)

3. Parameters (given data)

3a. Per good $g \in \mathcal{G}$ (data files)

Symbol	Meaning	Unit	Source
ε_g	energy per 100 g as stored	kcal / 100 g	nutrition.json (FAO)
π_g	protein per 100 g as stored	g / 100 g	nutrition.json
φ_g	fat per 100 g as stored	g / 100 g	nutrition.json
η_g	edible share	–	nutrition.json
δ_g	protein quality (DIAAS); defaults 1.0 animal, 0.44 plant	–	nutrition.json
n_{gi}	nutrient i per 100 g eaten	mg or µg / 100 g	nutrient_profiles.json (USDA SR Legacy, MEXT)
ϵ_g^{U}	energy per 100 g of the USDA form behind n_{gi}	kcal / 100 g	nutrient_profiles.json
ω_g	water per 100 g eaten	g / 100 g	nutrient_profiles.json
θ_g	sugars per 100 g	g / 100 g	USDA SR Legacy
α_g	alcohol per 100 g	g / 100 g	USDA SR Legacy
h_g	hydration index (drinks)	–	nutrition.json
m_g	one serving	g	USDA household portions
H_g	Holt satiety index, where measured (white bread = 100)	–	Holt et al. 1995
$\bar{\delta}_g$	spoilage of a year's leftover by keeping form	–	nutrition.json (FAO storage)

Symbol	Meaning	Unit	Source
$\ell_g,\ \chi_g$	brewing: litres of drink per kg; caffeine per litre ($g \in \mathcal{G}^{\mathrm{T}}$)	L/kg; mg/L	MEXT (43, 200 tea; 15, 600 coffee)

3b. Scalars

Symbol	Meaning	Value	Unit	Source
B	basal metabolic rate of the abstracted adult	1,536	kcal / (p·d)	FAO/WHO/UNU 2004; EFSA weights
Λ_k	activity level of work kind k	2.25, 1.76, 1.53	–	FAO 2004, Table 5.1
$\Lambda^{\min}, \Lambda^{\max}$	activity range	1.40, 2.40	–	FAO 2004, Table 5.3
κ	energy effect of caffeine at the dose γ	150	kcal / (p·d)	Dulloo et al. 1989
γ	caffeine dose for the full effect	600	mg / (p·d)	Dulloo et al. 1989
E^{ref}	reference energy for nutrient densities	2,600	kcal / (p·d)	nutrient_needs.json
r_i	daily need of nutrient i	per nutrient	mg, µg or g / (p·d)	EFSA 2017; Japanese DRI 2025 (sodium)
$\bar{\sigma}$	mean Holt index of Holt's 38 foods ÷ 100	1.36	–	Holt et al. 1995
λ	fullness factor of drinks	0.5	–	estimate (PMID 14649371)
$b_0, \ldots, b_4$	fitted satiety formula	0.0796, 0.2218, 0.6614, 0.0308, −0.2127	–	fitted to 29 Holt foods (`satiety_fit.py`)
$\kappa_s,\ \bar{p},\ \bar{\delta}$	protein factor: slope, bread's protein share, wheat's DIAAS	2.0, 0.146, 0.44	–	fitted to Holt's beef against bread
$u_1, \ldots, u_5$	well-being weights	1, 2, 1, 1, 0.5	–	estimates on sourced effects
β^{W}	water per kcal at which lightness is full	3	g / kcal	estimate
$\pi^{\max}$	protein ceiling, share of energy	0.35	–	Bilsborough and Mann 2006
$\Phi^{\max}$	fat from one good	250	g / (p·d)	the user (M93)
$Q^{\mathrm{milk}}, Q^{\mathrm{lemon}}, Q^{\mathrm{fruit}}, Q^{\mathrm{salt}}$	ceilings: milk, lemons, all fruit, salt	3,000; 116; 4,518; 10	g / (p·d)	the user; estimate; USDA watermelon; estimate
W^{need}	total water need	2,250	ml / (p·d)	EFSA 2010

Symbol	Meaning	Value	Unit	Source
a_0, a_1, a^{ref}	iron absorption: base, slope by animal protein share, EFSA's assumption	0.10, 0.08, 0.16	–	FAO/WHO 2004; EFSA 2017
Ω	ceiling of the work factor	1.20	–	estimate
Δ	step of the allocation	10	kcal	numerical

4. Variables

Symbol	Meaning	Domain	Unit
q_g	grams eaten per person a day	$g \in \mathcal{G}$, $q_g \geq 0$	g / (p·d)
E	energy eaten	$E \geq 0$	kcal / (p·d)
F	fullness eaten	$F \geq 0$	fullness units (white-bread kcal) / (p·d)
P	protein eaten	$P \geq 0$	g / (p·d)
R	fruit eaten	$R \geq 0$	g / (p·d)

5. Equations

M1. Availability

$$a_g = \frac{S_g \cdot 10^6}{365\,N} \qquad g \in \mathcal{G} \tag{1}$$

M2. Need and bounds

$$\begin{aligned} q_g &= \min\Big(a_g,\ \frac{1000\,\big(\gamma - \sum_{g' < g} q_{g'} \ell_{g'} \chi_{g'} / 1000\big)}{\ell_g \chi_g}\Big) \qquad g \in \mathcal{G}^{\mathrm{T}} \\ c &= \min\Big(1,\ \frac{1}{\gamma} \sum_{g \in \mathcal{G}^{\mathrm{T}}} q_g\, \frac{\ell_g \chi_g}{1000}\Big) \end{aligned} \tag{2}$$

$$\Lambda^{\mathrm{w}} = \sum_{k \in \mathcal{K}} w_k \Lambda_k \tag{3}$$

$$E^{\text{need}} = B\,\Lambda^{\mathrm{w}} + \kappa c \tag{4}$$

$$E^{\max} = B\,\Lambda^{\max} + \kappa c \tag{5}$$

$$F^{\text{fed}} = \bar{\sigma} E^{\text{need}}, \qquad F^{\max} = \bar{\sigma} E^{\max} \tag{6}$$

$$P^{\max} = \frac{\pi^{\max} E^{\max}}{4} \tag{7}$$

$$q_{\text{salt}} = \min(a_{\text{salt}}, Q^{\text{salt}}) \tag{8}$$

M3. Indicators per good $g \in \mathcal{G}^{\text{E}}$

Energy, protein and fat per gram eaten:

$$e_g = \frac{\varepsilon_g \eta_g}{100}, \qquad \hat{\pi}_g = \frac{\pi_g \eta_g}{100}, \qquad \hat{\varphi}_g = \frac{\varphi_g \eta_g}{100} \tag{9}$$

Energy shares:

$$p_g = \frac{4\pi_g}{\varepsilon_g}, \quad f_g = \min\Big(1, \frac{9\varphi_g}{\varepsilon_g}\Big), \quad l_g = \frac{7\alpha_g}{\varepsilon_g}, \quad k_g = \max(0,\, 1 - f_g - p_g - l_g), \quad z_g = \frac{4\theta_g}{\varepsilon_g} \tag{10}$$

Fullness per kcal:

$$\sigma_g = \Big[\frac{H_g}{100} \text{ if measured, else } \exp\big(b_0 + b_1 \ln \tfrac{W_g}{W_{\text{bread}}} + b_2 p_g + b_3\, \text{fib}_g + b_4 f_g\big)\Big] \cdot \begin{cases} \lambda & g \in \mathcal{G}^{\text{D}} \\ 1 & \text{otherwise} \end{cases} \tag{11}$$

Here W_g is the grams in 240 kcal and fib_g the fibre in g per 100 kcal.

Satisfaction (protein factor):

$$s_g = \frac{1 + \kappa_s\, p_g \min(\delta_g, 1)}{1 + \kappa_s\, \bar{p}\, \bar{\delta}} \tag{12}$$

Nutrient density against the requirement density:

$$\nu_g = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \min\Big(1,\ \frac{n_{gi}/\epsilon^{\text{U}}_g}{r_i / E^{\text{ref}}}\Big) \tag{13}$$

For protein, $n_{gi}/\epsilon^{\text{U}}_g$ is replaced by π_g/ε_g.

Well-being:

$$\beta_g = u_1 z_g + u_2 f_g k_g + u_3 l_g + u_4 \min\Big(1, \frac{\omega_g/\epsilon^{\text{U}}_g}{\beta^{\text{W}}}\Big) - u_5 f_g \tag{14}$$

The two keys (two separate sorted lists, M96):

$$K^{(1)}_g = s_g + \nu_g \quad \text{(nutrition)}, \qquad K^{(2)}_g = \beta_g \quad \text{(well-being)} \tag{15}$$

The ceiling of each good:

$$\bar{q}_g = \min\Big(a_g,\ \frac{\Phi^{\max}}{\hat{\varphi}_g},\ Q^{\text{milk}}\,[g = \text{Dairy}],\ Q^{\text{lemon}}\,[g = \text{Lemon}]\Big) \tag{16}$$

A bracketed condition means the term applies only to that good.

M4. Allocation

The marginal value of the next Δ kcal of good g in pass ρ falls with what has been eaten:

$$\mu_g^{(\rho)}(q_g) = \frac{K_g^{(\rho)}}{1 + q_g / m_g} \tag{17}$$

A good is admissible when the next step keeps it within its own ceiling, the protein ceiling and the fruit ceiling:

$$\mathcal{A}(\mathbf{q}) = \Big\{ g \in \mathcal{G}^{\mathrm{E}} :\; q_g + \tfrac{\Delta}{e_g} \le \bar{q}_g,\;\; P + \tfrac{\Delta}{e_g}\hat{\pi}_g \le P^{\max},\;\; [g \in \mathcal{G}^{\mathrm{F}}] \Rightarrow R + \tfrac{\Delta}{e_g}\eta_g \le Q^{\mathrm{fruit}} \Big\} \tag{18}$$

Algorithm 1 (allocation):

```
q_g ← 0 for all g ∈ G^E;  E ← 0;  F ← 0;  P ← 0;  R ← 0
for ρ in (1, 2):
    F^stop ← F^fed if ρ = 1 else F^max
    while E < E^max and F < F^stop:
        A* ← { g ∈ A(q) : K^(ρ)_g > 0 }
        if A* is empty: break
        g* ← argmax over g ∈ A* of μ^(ρ)_g(q_g)
        q_g* ← q_g* + Δ / e_g*
        E ← E + Δ;   F ← F + Δ σ_g*;   P ← P + (Δ / e_g*) π̂_g*
        if g* ∈ G^F: R ← R + (Δ / e_g*) η_g*
return q, E, F, P
```

M5. Outcomes

$$fed = \frac{F}{F^{\mathrm{fed}}}, \qquad x = \frac{E}{E^{\mathrm{need}}} \tag{19}$$

$$I_i = \sum_{g \in \mathcal{G}} \frac{q_g \eta_g}{100}\, n_{gi} \quad (\text{for } g \in \mathcal{G}^{\mathrm{T}},\; q_g\eta_g \text{ is replaced by } q_g \ell_g) \tag{20}$$

$$a^{\mathrm{Fe}} = a_0 + a_1 \frac{\sum_{g \in \mathcal{G}^{\mathrm{A}}} q_g \hat{\pi}_g}{P} \tag{21}$$

$$c_i = \frac{I_i}{r_i} \cdot \begin{cases} a^{\mathrm{Fe}} / a^{\mathrm{ref}} & i = \text{iron} \\ 1 & \text{otherwise} \end{cases} \tag{22}$$

$$W^{\mathrm{food}} = \sum_{g \notin \mathcal{G}^{\mathrm{D}}} \frac{q_g \eta_g}{100}\, \omega_g, \qquad W^{\mathrm{drink}} = \sum_{g \in \mathcal{G}^{\mathrm{D}} \setminus \mathcal{G}^{\mathrm{T}}} q_g \eta_g h_g + \sum_{g \in \mathcal{G}^{\mathrm{T}}} q_g \ell_g \tag{23}$$

$$W^{\mathrm{plain}} = \max\!\big(0,\; W^{\mathrm{need}} - W^{\mathrm{food}} - W^{\mathrm{drink}}\big) \tag{24}$$

$$\Lambda^{\mathrm{act}} = \min\!\big(\Lambda^{\max}, \max(\Lambda^{\min}, E/B)\big) \tag{25}$$

$$work = \min\!\Big(\Omega, \max\!\big(0, \tfrac{\Lambda^{\mathrm{act}} - \Lambda^{\min}}{\Lambda^{\mathrm{w}} - \Lambda^{\min}}\big)\Big) \tag{26}$$

$$U_g = \frac{365\, N\, q_g}{10^6}, \qquad L_g = S_g - U_g, \qquad D_g = \bar{\delta}_g L_g \tag{27}$$

The Fish, Cow and Pig goods are composites (whole animals). Their $\varepsilon, \pi, \varphi, n_{gi}$ are weighted means of their parts:

$$\varepsilon_g = \sum_j \xi_{gj} \varepsilon_{gj}, \;\ldots \qquad \text{Fish: } \xi = (0.50 \text{ herring},\; 0.45 \text{ cod},\; 0.05 \text{ liver});\; \text{Cow, Pig: } (0.97 \text{ meat},\; 0.03 \text{ liver}) \tag{28}$$

6. Properties

(P1) Energy at a fullness stop. From Algorithm 1, line 9, $F = \sum_g \sigma_g\, e_g q_g$ and $E = \sum_g e_g q_g$. When eating stops because F reached F^{stop}:

$$E = \frac{F^{\text{stop}}}{\tilde{\sigma}}, \qquad \tilde{\sigma} = \frac{\sum_g \sigma_g\, e_g q_g}{\sum_g e_g q_g}, \tag{29}$$

the energy-weighted mean fullness of the diet. Filling diets give less energy at the same fullness.

(P2) What each pass optimises. The marginal value (17) is the derivative, per kcal, of

$$V_g^{(\rho)}(q_g) = K_g^{(\rho)}\, e_g\, m_g \ln\!\Big(1 + \frac{q_g}{m_g}\Big). \tag{30}$$

Taking always the largest marginal value in equal energy steps is the greedy method for a separable concave objective under one linear budget. So each pass approximates, to within one step Δ:

$$\max_{\mathbf{q}} \sum_g V_g^{(\rho)}(q_g) \quad \text{subject to} \quad \sum_g e_g q_g \le E^{(\rho)},\;\; q_g \le \bar{q}_g,\;\; P \le P^{\max},\;\; R \le Q^{\text{fruit}},$$

where $E^{(\rho)}$ is the energy at which the pass stops (fullness or $E^{\max}$).

(P3) Step overshoot. F and E can exceed their stop by at most one step: $\Delta \max_g \sigma_g$ and Δ.

7. What the outcomes feed (proposed, M93)

Outcome	Feeds
fed	fertility and the famine term of the population model
x, $work$	the weakness of the underfed; physical output
c_i	the deficiency burden, the nutrients factor of fertility, output
W^{plain}	demand on the water supply (D31)
L_g, D_g	stock, storage, trade, recipes

8. Change log

Version (round)	Variable	Change	Before → after	Reason
1 (M85)	E^{need}	changed	2,600 kcal → (4)	the user: activity and stimulants
1 (M86)	$E^{\max}$	added	→ (5)	FAO PAL ceiling
1 (M86)	x	changed	satiety ÷ 2,600 → E/E^{need} (19)	need varies with work
1 (M86)	$work$, Ω	added	→ (26)	Fogel
1 (M87)	a_g	added	→ (1)	the goods pass counts tonnes
1 (M87)	c_i, a^{Fe}	added	→ (20)-(22)	deficiency model

Version (round)	Variable	Change	Before → after	Reason
1 (M88)	headroom, food mass, drink cap	removed	0.15, 6 kg, 3.5 L → none	the user
1 (M88)	Q^{fruit}; brewing $\ell_g, \chi_g; \gamma$	added	→ 4,518 g; MEXT; 600 mg	the user
1 (M91)	W^{need}, (23)-(24)	changed	drink 1.8 L → total water 2,250 ml	the user: one hydration equation
1 (M91)	m_g, (17)	added	→ diminishing returns per serving	the user: no tiers
1 (M91)	σ_g formula, λ	added	→ (11)	the user: a correct formula
1 (M91)	β_g	added	→ (14)	the user: well-being
1 (M91)	Fish	changed	fillet → composite (28)	the user: whole animal
1 (M92)	$F^{\max}$, $\bar{\sigma}$; Q^{lemon}	added	→ (6); 116 g	the user: fullness; lemons
1 (M93)	$\Phi^{\max}$	changed	100 → 250 g	the user
1 (M93)	fed, F^{fed}	added	→ (6), (19)	the user: satiety is the baseline
1 (M94)	ν_g	changed	per 100 kcal ÷ daily need → (13)	scaling error (beef 0.07 → 0.71)
1 (M94)	Cow, Pig	changed	muscle → composite with liver (28)	the user
2 (M96)	preference $V_g = s_g + \nu_g + \beta_g$	removed	one key → $K^{(1)}, K^{(2)}$ (15)	the user: two separate variables and sorted lists
2 (M96)	allocation	changed	one pass → two passes (Algorithm 1)	M88: nutrition first, then well-being
3 (M98)	—	presentation only	prose and code-style formulas → sets, parameters, variables, equations (1)-(30), Algorithm 1, legend	the user: proper notation and a full legend

9. Rules for changes

1. A change is written here first, in sections 2-5, with a row in section 8.
2. The code changes only what the row names; the commit names the row.
3. Results are published with the version they come from.